

B.TECH.- Third SEMESTER
MID-SEMESTER EXAMINATION, September 2024

Course Code-CMMTC303

Course Title-Applied Linear Algebra

Time: 1:30 hours

Max Marks-20

Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1(a)	Check whether the set of positive rationals with $a * b = ab/2$ is a group or not, where $a, b \in \mathbb{Q}^+$.	2	CO1
1(b)	Show that the set $\mathbb{R}^2$ is a vector space over $\mathbb{R}$.	2	CO1
2(a)	Let V be the space of functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Check if $W = \{f(x) : f(1) = 0\}$ is a subspace of V or not.	2	CO1
2(b)	Show that $W_1 \oplus W_2 = \mathbb{R}^2$, where $W_1 = \{(x, 2x) : x \in \mathbb{R}\}$ and $W_2 = \{(x, 3x) : x \in \mathbb{R}\}$.	2	CO1
3(a)	Is the set A linearly independent in $\mathbb{R}^4$? $A = \{(1, 1, 2, 4), (2, -1, -5, 2), (1, -1, -4, 0), (2, 1, 1, 6)\}$.	2	CO1
3(b)	Let $U = \{(x, y, z) : 2x + 3y + z = 0\}$ and $V = \{(x, y, z) : x + 2y - z = 0\}$ be subspaces of $\mathbb{R}^3$. Find bases for U, V and $U \cap V$.	2	CO1
4(a)	Verify rank-nullity theorem for the linear transformation $T : P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ such that $T(f(x)) = xf(x) + f'(x)$.	2	CO2
4(b)	For a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x + 2y - 3z, 2x + 5y - 4z, x + 4y + z)$, find a basis of image of T .	2	CO2

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5(a)	Show that the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(x, y) = (x + y, x)$ is invertible.	2	CO2
5(b)	Is there a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ such that $T(1, 0, 3) = (1, 1)$ and $T(-2, 0, -6) = (2, 1)$?	2	CO2

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